

Fig. 1: Sekisui Chemical Co. Ltd. - production characteristics: company-wide MFCA implementation for 34 sites with individually-different production characteristics

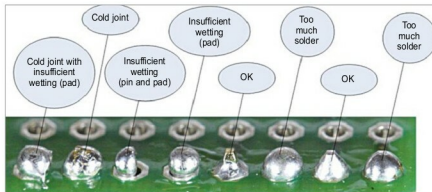


Fig. 2: An example of poor soldering

be answered and documented during design phases are:

- What is the anticipated number of plan modifications that may happen during the development stage?
 - How many elements should be involved during plan modification?
 - How much rework effort is required for plan modification?
 - Which component requires optimal rework effort?
 - Which ingredients should be prepared early to minimise shock due to design modifications?
- Database structure that may be useful to analyse and foresee design change efforts is given below:
- Unique design change number
 - Description of design change
 - Name of module or component where design change needs to be made
 - Type of change
 - Time spent to carry out the change
 - Name(s) of person(s) who made the change

- Date on which design change is made

Cost of material waste. Any material/tool that is unavailable due to defect, mishandling or being scrapped is material waste. In every organisation, during execution of its process there is always some loss incurred. Material loss margin varies from company to company, depending on how they are checking it. A few material losses can be manipulated effectively, whereas others cannot.

Especially in the research and development sector, loss due to bad workmanship such as test equipment error, development/test setup issue, choosing right vendors for quality service and materials, and the like can be curbed by training, whereas material losses during the making of prototypes are inevitable.

By preventing material waste, costs can be spared in any kind of establishment. A few examples where industries took in huge costs towards material waste are given in

"For every customer complaint there are 26 other unhappy customers who have remained silent."

—Lee Resource

Tables I and II, and Fig. 1.

Waste material can be imaged in consumer electronics based industries in the following ways:

- When defective tools, equipment or raw materials (like spare parts and consumables) are purchased
- Test equipment errors due to lack in regular calibrations
- Mishandling of equipment/poor workmanship
- Bad-quality materials
- Wrong settings or calibrations
- Poor inspection
- Lack of environmental controls
- Poor supervision of workmen

Cost due to non-conformance in manufacturing process. Non-conformance in the manufacturing process is an event that deviates from certain manufacturing protocols and criteria. Usually, manufacturing process management experts are keen to curb these losses. Non-conformance management solutions enable identification and documentation of these quality events, apply standard risk criteria to triage, route events appropriately and enforce structured failure analysis, root cause identification, quarantine and final disposition.

There are various factors in the manufacturing process that may incur as contribution to manufacturing defects. Some factors that are observed as part of the manufacturing process are mentioned below:

- Lack in availability or usage of electrostatic discharge (ESD) devices like ESD wristbands, ESD tables, ESD clothes and the like
- Human hair that may fall on PCBs/components in assembly lines and cause damage to the board due to ESD generated from hair
- Thermal balance
- PCB cracked or broken dur-